Acetic Acid Content in Vinegar

The acetic acid content is determined in vinegar by titration with sodium hydroxide.

Sample	Vinegar, 0.7-1 g.
Compound	Acetic acid, CH_3COOH, M = 60.05 g/mol, z = 1
Chemicals	50 mL deionized water
Titrant	Sodium hydroxide, NaOH c(NaOH) = 0.1 mol/L
Standard	Potassium hydrogen phthalate, KHP, 80 mg
Indication	DG115-SC
Chemistry	$CH_3COOH + NaOH \rightarrow$ $CH_3COONa + H_2O$
Calculation	Content (%), expressed as w/w: R1 = Q*C/m, C = M/(10*z)
Waste disposal	Neutralization before final disposal as aqueous solution.
Author, Version	Thomas Hitz, MSG, July 2006 Ralph Egli, MSG, May 2009

Preparation and Procedures

Note:

This method allows a fully automated analysis procedure. The method can be easily modified for manual operation. Select "Manual stand" in the method function "Titration stand".

1) Adjustment of the DG115-SC pH glass electrode: The electrode is adjusted using pH buffers 4.01, 7.00, and 9.21 before starting the analysis.
2) The titer of sodium hydroxide is determined using potassium hydrogen phthalate as a primary standard (KHP): 80 mg is dissolved into 50 mL deionized water and titrated.
3) Approximately 1 g vinegar is added to 50 mL deionized water into a titration beaker.
4) In the T50/T70/T90 application, the sample series was analyZed using a Rondolino sample changer in combination with a diaphragm pump.

   The rinsing time was defined to 2 s. In this way, the electrode is cleaned with deionized water before starting the subsequent sample.

Remarks

1) The method parameters have been developed and optimized for the above mentioned sample.
2) Since there are different kinds of vinegar, it may be necessary to slightly adapt the method to your specific sample.

Instruments	
	- T50/T70/T90 Titration Excellence, G20 Compact Titrator (with small changes). - XS205 Balance - Rondolino Sample Changer with diaphragm pump and PowerShower™

Accessories	
	- 1 x 10 mL DV1010 burette - Titration beaker ME-101974 - Olivetti Printer JobJet 210

Results T50/T70/T90

All results

Method-ID	M400
Sample	Vinegar (1/1)
R1 (Content)	4.713 %
Sample	Vinegar (1/2)
R1 (Content)	4.734 %
Sample	Vinegar (1/3)
R1 (Content)	4.718 %
Sample	Vinegar (1/4)
R1 (Content)	4.717 %
Sample	Vinegar (1/5)
R1 (Content)	4.722 %
Sample	Vinegar (1/6)
R1 (Content)	4.724 %

Statistics

Method-ID	M400
R1	Content
Samples	6
Mean	4.738 %
s	0.007 %
srel	0.152 %

Titration curve T50/T70/T90

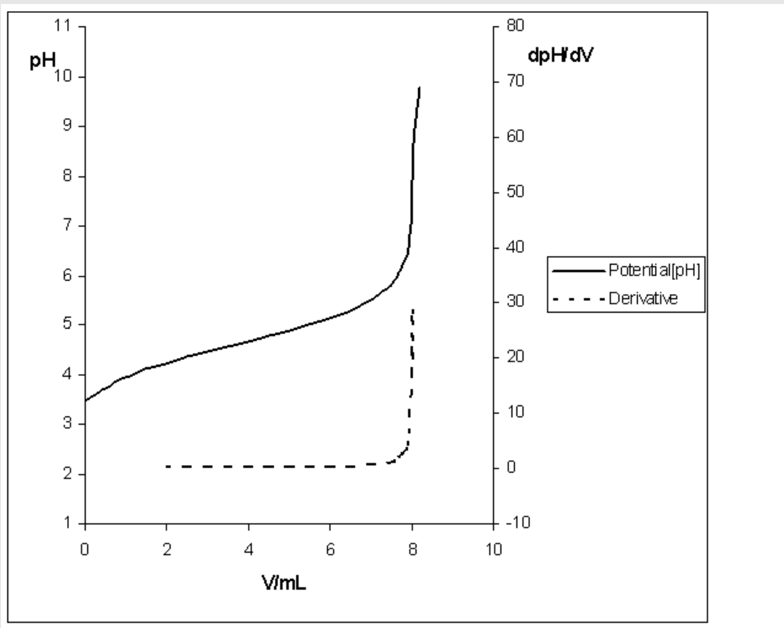

Table of measured values T50/T70/T90

Consumption / mL	SensorValue / mV	Potential / pH	Derivative / dpH/dV	Time / s
0	209.1	3.466	NaN	0
0.571	191	3.771	NaN	3.9
0.857	183.8	3.892	NaN	7
1	180.5	3.948	NaN	10.1
1.5	171.1	4.107	NaN	18.7
2	163.1	4.243	0.25	22
2.5	156.1	4.362	0.23	25.3
3	149.5	4.473	0.21	28.9
3.5	143.3	4.578	0.21	31.9
4	137.1	4.682	0.2	35.2
4.5	131	4.786	0.21	38.4
5	124.6	4.894	0.21	41.8
5.5	117.7	5.01	0.23	44.9
6	109.9	5.143	0.29	48.4
6.5	100.8	5.297	0.4	51.5
7	88.6	5.502	0.66	54.8
7.484	70	5.817	1.28	58.8
7.682	56.7	6.042	1.96	62.8
7.784	46.6	6.212	2.76	65.9
7.866	34.6	6.416	4	69
7.922	21.1	6.643	6.42	73.2
7.952	10	6.831	9.74	77.2
7.972	-1.2	7.02	13.41	81.6
7.986	-11.9	7.2	17.22	86.8
7.998	-24	7.405	21.07	92.4
8.007	-36.4	7.616	23.47	98.7
8.014	-47.6	7.805	26.48	105.2
8.02	-60	8.014	28.98	112.2
8.024	-68.6	8.16	26.76	118.7
8.03	-77.8	8.315	23.97	125.1
8.041	-93.2	8.575	20.59	131.5
8.05	-103.1	8.742	17.81	136.9
8.064	-114	8.927	NaN	141.5
8.086	-126.4	9.136	NaN	146
8.116	-138.6	9.343	NaN	150.3
8.157	-151.5	9.561	NaN	154.4
8.206	-164	9.772	NaN	158.7

Comments T50/T70/T90

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Results G20

All results

Method-ID	M400
Sample	Vinegar (1/6)
R1 (Content)	4.781 %
Sample	Vinegar (2/6)
R1 (Content)	4.763 %
Sample	Vinegar (3/6)
R1 (Content)	4.776 %
Sample	Vinegar (4/6)
R1 (Content)	4.779 %
Sample	Vinegar (5/6)
R1 (Content)	4.780 %
Sample	Vinegar (6/6)
R1 (Content)	4.774 %

Statistics

Method-ID	M400
R1	Content
Samples	6
Mean	4.776 %
s	0.007 %
srel	0.139 %

Titration curve G20

Not available

Table of measured values G20

Not available

Comments G20

1. For the G20 Compact Titrator results, no automation was used. Each sample has been added manually to the titration stand.

Method T50/T70/T90

001 Title

Type	General titration
Compatible with	T50 / T70 / T90
ID	M400
Title	Acetic acid content in vinegar
Author	METTLER TOLEDO
Date/Time	02.08.2006 15:00:00
Modified at	–
Modified by	–
Protect	No
SOP	None

002 Sample

Number of IDs	1
ID 1	Vinegar
Entry type	Weight
Lower limit	0.7 g
Upper limit	1.0 g
Density	1.0 g/mL
Correction factor	1.0
Temperature	25.0°C

003 Titration stand (Rondolino TTL)

Type	Rondolino TTL
Titration stand	Rondolino TTL 1

004 Stir

Speed	35%
Duration	10 s
Condition	No

005 Titration (EQP) [1]

Titrant	
Titrant	NaOH
Concentration	0.1 mol/L
Sensor	
Type	pH
Sensor	DG115-SC
Unit	pH
Temperature acquisition	
Temperature acquisition	No
Stir	
Speed	35%
Predispense	
Mode	Volume
Volume	1 ml
Wait time	5 s
Control	
Control	User
Titrant addition	Dynamic
dE (set value)	12 mV
dV (min)	0.005 mL
dV (max)	0.5 mL
Meas. val. acquisition	Equilibrium controlled
dE	0.5 mV
dt	1 s
t (min)	3 s
t (max)	30 s
Evaluation and recognition	
Procedure	Standard
Threshold	5 pH/mL
Tendency	Positive
Ranges	0
Add. EQP criteria	No
Termination	
At Vmax	10.0 mL
At potential	No
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination criteria	No

006 Calculation R1

Result	Content
Result unit	%
Formula	R1=Q*C/m
Constant C	M/(10*z)
M	M[Acetic acid]
z	z[Acetic acid]
Decimal places	3
Result limits	No
Record statistics	Yes
Extra statistical func.	No
Send to buffer	No
Condition	No

007 Record

Results	Per series
Raw results	Per series
Table of meas. values	Last titration function
Sample data	Per series
Resource data	No
E - V	Last titration function
dE/dV - V	Last titration function
log dE/dV - V	No
d^2E/dV^2 - V	No
BETA – V	No
E - t	No
V - t	No
dV/dt - t	No
T – t	No
E – V & dE/dV – V	No
V – t & dV/dt – t	No
Method	No
Series data	No
Condition	No

008 End of sample

Method G20 Compact Titrator

001 Title

Type	General titration
Compatible with	G20
ID	M400
Title	Acetic Acid in Vinegar
Author	Administrator
Date/Time	02.08.2009 15:00:00
Modified at	–
Modified by	–
Protect	No
SOP	None

002 Sample

Number of IDs	1
ID 1	Vinegar
Entry type	Weight
Lower limit [g]	0.7
Upper limit [g]	1.0
Density [g/mL]	1.0
Correction factor	1.0
Temperature [°C]	25.0
Entry	Arbitrary

003 Titration stand (Rondolino TTL)

Type	Rondolino TTL
Titration stand	Rondolino TTL 1

004 Stir

Speed [%]	35
Duration [s]	10

005 Titration (EQP) [1]

Titrant	
Titrant	NaOH
Concentration [mol/L]	0.1
Sensor	
Type	pH
Sensor	DG115-SC
Unit	pH
Temperature acquisition	
Temperature acquisition	No
Stir	
Speed [%]	35
Predispense	
Mode	Volume
Volume [mL]	1.0
Wait time [s]	5
Control	
Control	User
Titrant addition	Dynamic
dE (set value) [mV]	12.0
dV (min) [mL]	0.005
dV (max) [mL]	0.5
Mode	Equilibrium controlled
dE [mV]	0.5
dt [s]	1
t (min) [s]	3
t (max) [s]	30
Evaluation and recognition	
Procedure	Standard
Threshold [pH/mL]	5
Tendency	Positive
Ranges	0
Add. EQP criteria	No
Termination	
At Vmax [mL]	20.0
At potential	No
At slope	No
After number of recognized EQPs	Yes
Number of EQPs	1
Combined termination criteria	No

006 Calculation R1

Result type	Predefined
Calculation type	Direct titration
Result	Content
Result unit	%
Formula	R1=Q*C/m
Selected EQP	1
Constant	C=M/(10*z)
M	M[Acetic acid]
z	z[Acetic acid]
Decimal places	3
Result limits	No
Record statistics	Yes

007 Record

Summary	No
Results	Per series
Raw results	Per series
Table of meas. values	Last titration function
Sample data	Per series
Resource data	No
E - V	Last titration function
dE/dV - V	Last titration function
log dE/dV - V	No
d^2E/dV^2 - V	No
BETA – V	No
E - t	No
V - t	No
dV/dt - t	No
T – t	No
E – V & dE/dV – V	No
V – t & dV/dt – t	No
Method	No
Series data	No
Condition	No

008 End of sample